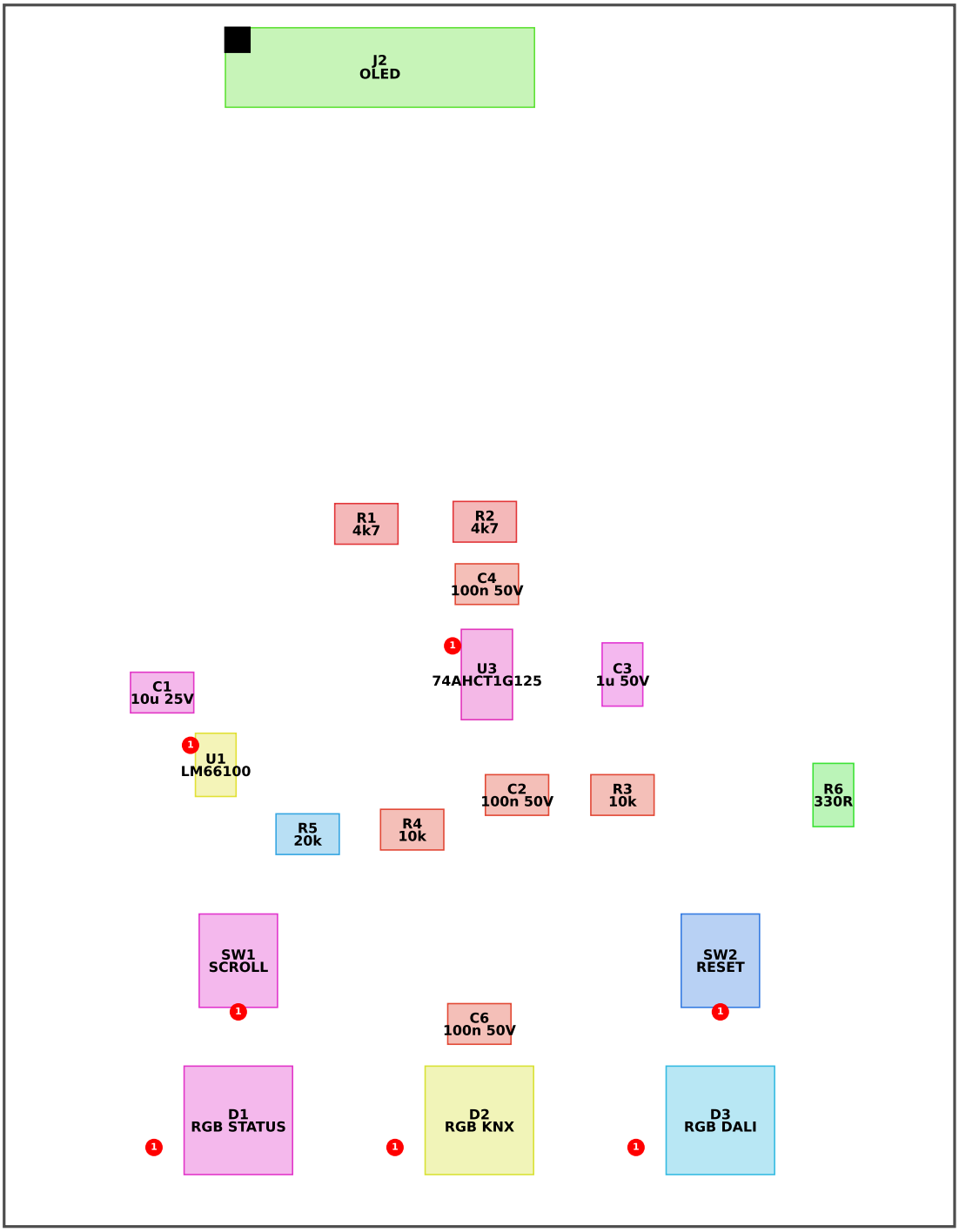
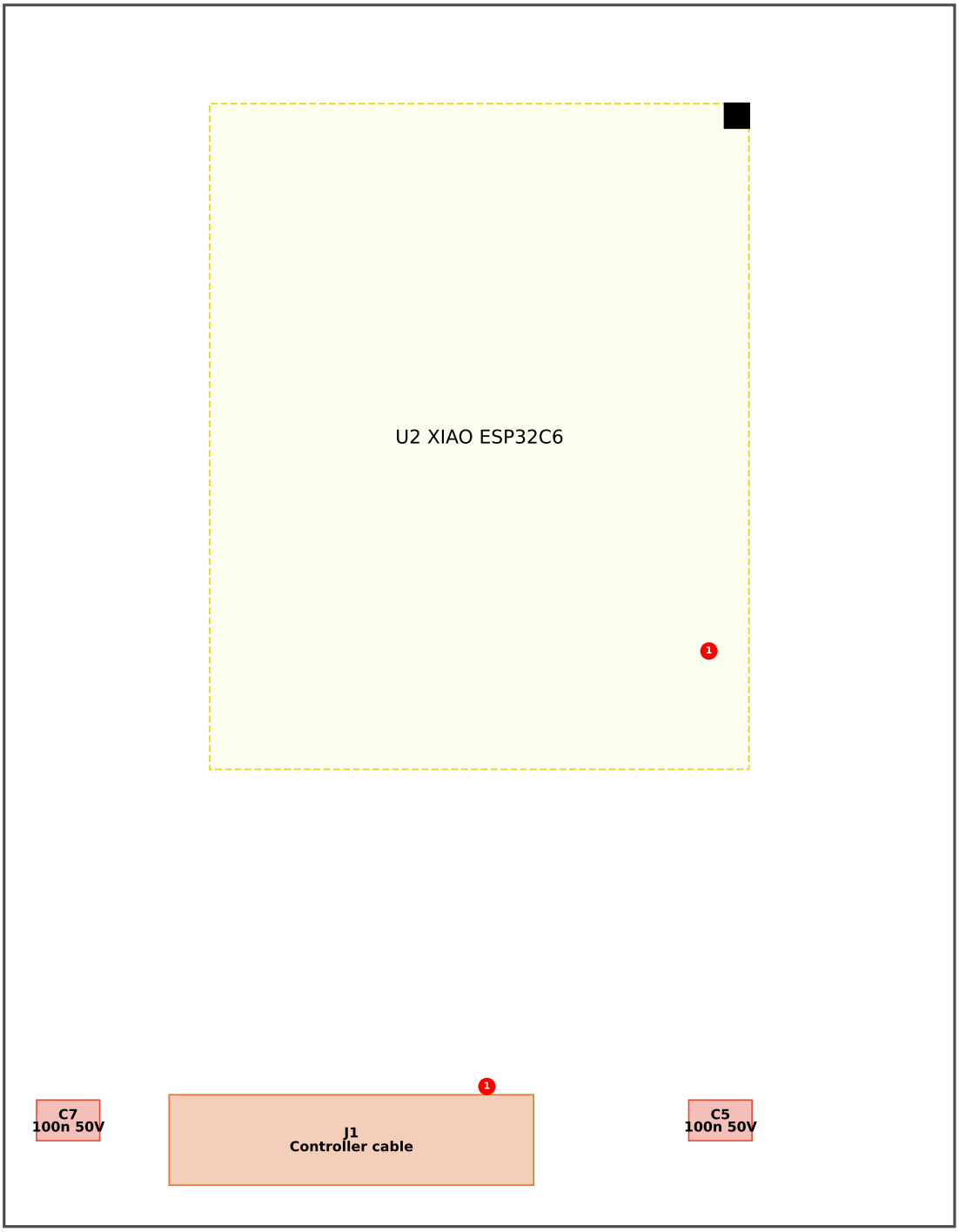
ui - top side

black bar = diode cathode, "+" = electrolytic +, red "1" = pin 1, no mark = either way;
black corner = hand-fitted, dashed = large part, colour = value (key on the last pages)



ui - bottom side (mirrored, seen from below)
black bar = diode cathode, "+" = electrolytic +, red "1" = pin 1, no mark = either way;
black corner = hand-fitted, dashed = large part, colour = value (key on the last pages)



ui - values (23 parts)

100n 50V	x2	bottom	C5, C7
Controller cable	x1	bottom	J1
XIAO ESP32C6	x1	bottom	U2
100n 50V	x3	top	C2, C4, C6
10k	x2	top	R3, R4
10u 25V	x1	top	C1
1u 50V	x1	top	C3
20k	x1	top	R5
330R	x1	top	R6
4k7	x2	top	R1, R2
74AHCT1G125	x1	top	U3
LM66100	x1	top	U1
OLED	x1	top	J2
RESET	x1	top	SW2
RGB DALI	x1	top	D3
RGB KNX	x1	top	D2
RGB STATUS	x1	top	D1
SCROLL	x1	top	SW1

ui - how to read this drawing and the parts

Black bar = diode cathode (K). The diode body carries a band on that end.

"+" = electrolytic capacitor + pad. The can has a dark stripe (or "-" marks) on the other side.

Red dot "1" = pin 1 of ICs, transistors, LEDs, optocouplers, connectors and modules:

- SOT-23 / SOT-23-5 / TSOT-23-6 / SC-70-6: pin 1 is marked by a dot or bevel, or follow the leg pattern (SOT-23: the single leg is pin 3, opposite pins 1 and 2).
- SO-6 optocouplers, the NCN5130 (QFN) and the supervisor: a dot or notch at pin 1 on the package.
- WS2812B-MINI-V3J: a notch / marked corner on the body is pin 1 (VDD); the legs differ, so the footprint only fits one way.
- The XIAO, DC-DC converters and the REC3 module: their silkscreen / pin-1 marking and the socket shape.

No mark = fits either way (resistors, ceramic capacitors, inductors, fuse, crystal, pin strips).

Black corner = fitted by hand (not placed by the assembler); dashed outline = large part.

Same colour = same value (see the values page).

Tools: a 3-5x head magnifier or a 10x loupe and good light are enough to read the bands and pin-1 dots on SOT-23 and 0805 parts; a USB microscope helps for the NCN5130 QFN and SC-70-6.